

RAJAN TANDUKAR

Senior Software Engineer | Technical Leader | Kathmandu, Nepal | +977 9841195643 | royal_raj@outlook.com | AWS Certified Developer Associate

PROFESSIONAL SUMMARY

Senior Software Engineer and Technical Leader with **13+ years of experience** designing and delivering scalable enterprise SaaS, cloud-native, and distributed systems. Strong expertise in **software architecture, microservices, .NET, Node.js, React, AWS, PostgreSQL, and event-driven systems**. Experienced in evaluating architectural options and trade-offs, translating complex business requirements into scalable technical solutions, and providing technical leadership through architecture reviews, code reviews, peer programming, and mentoring. Hands-on experience building **LLM-powered AI applications, RAG pipelines, vector database solutions, semantic search, and AI-assisted software engineering workflows** using **Claude Code, GitHub Copilot, and OpenAI Codex**.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, SQL

Backend & Architecture: .NET, ASP.NET Core, Web API, EF Core, Dapper, Node.js, NestJS, Microservices, REST APIs, Event-Driven Architecture, Distributed Systems, Clean Architecture, System Design

AI & AI-Assisted Engineering: LLMs, Generative AI, RAG, Vector Databases, Vector Embeddings, Semantic Search, AI-Powered Applications, Claude Code, GitHub Copilot, OpenAI Codex

Cloud & DevOps: AWS, Lambda, EC2, API Gateway, SQS, SNS, Cognito, SES, DynamoDB, CloudFormation, Azure, Azure DevOps, GitHub Actions, Docker, CI/CD

Databases: PostgreSQL, SQL Server, MongoDB, DynamoDB

Engineering Practices: Architecture Reviews, Code Reviews, Unit Testing, Peer Programming, Agile/Scrum, Technical Leadership, Mentoring, Performance Optimization

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Avionté Staffing Software | Apr 2024 - Present

- Provide **technical leadership and architectural guidance** for enterprise staffing and recruiting platforms, driving scalable, secure, maintainable solutions aligned with business objectives.
- Design and implement **microservices-based architectures** and scalable backend services using modern .NET technologies, defining service boundaries, API contracts, integration patterns, and data flows.
- Analyze complex and ambiguous technical problems, evaluate **multiple architectural options and trade-offs** across scalability, performance, security, cost, maintainability, and time-to-market, and recommend appropriate solutions.
- Designed and implemented **LLM-powered AI chat and RAG** solutions using vector databases, embeddings, semantic search, and contextual retrieval.
- Leverage **Claude Code, GitHub Copilot, and OpenAI Codex** for code generation, refactoring, debugging, test development, code analysis, documentation, and delivery acceleration while validating quality, security, and architectural impact.
- Lead **architecture and design reviews, code reviews, and peer programming sessions**, promoting engineering standards and maintainable, scalable solutions.
- Develop and maintain **unit tests** to improve code quality, reliability, and regression coverage.
- Mentor **junior and mid-level engineers** on system design, implementation, debugging, testing, architecture, and engineering practices.
- Collaborate with product managers, architects, QA engineers, and stakeholders to translate **complex business requirements into scalable technical solutions**.
- Participate in **Agile sprint planning, retrospectives, backlog discussions, and technical planning**.
- Evaluate technologies and architectural patterns, identify technical risks, and drive engineering decisions aligned with long-term product goals.

Senior Software Engineer | AzrachIT | 2022 - Apr 2024

- Rewrote legacy .NET Framework applications into Node.js (NestJS) and React solutions.
- Implemented Azure DevOps CI/CD pipelines and AWS deployments using Elastic Beanstalk and EC2.
- Mentored junior developers and collaborated with project managers on feature delivery.
- Key Projects: TradeSlot Auction System, CleanCog NetZero Engagement Platform.

Software Engineer | Optimatech, Melbourne | 2021-2022

- Developed cloud-native SaaS applications using .NET, React, and AWS services.
- Worked with Lambda, SQS, SNS, CloudFormation, Amplify, and serverless architectures.
- Delivered multi-tenant energy and expense management solutions.

Software Engineer | Consoltica | 2020-2021

- Built web APIs and enterprise applications using ASP.NET Core and React.
- Maintained inventory, ticketing, asset management, and personnel tracking platforms.

Software Engineer | IMS Software | 2019-2020

- Developed desktop, web, and mobile applications using WPF, Xamarin, Angular, and .NET Core.
- Delivered ticketing, POS, kiosk, and parking management systems.

Software Engineer | Elite Solutions | 2017-2019

- Developed banking and financial software solutions including reconciliation systems and self-service kiosks.
- Built applications using ASP.NET MVC and WPF.
- Provided on-site technical support to clients, troubleshooting business-critical applications.
- Conducted on-site client consultations and requirements gathering, translating business needs into technical requirements.

Web Developer | Networking Professionals of Nepal | 2013-2017

- Engineered an internal CSS template generation framework that automated repetitive UI development tasks and enforced uniform styling standards across corporate web initiatives.

EDUCATION

BSc (Hons) Computing - Islington College, Nepal

CERTIFICATIONS

- AWS Certified Developer - Associate